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CaML-OP: Causal Machine Learning for Oncology Personalization

A Hybrid Predictive–Causal Framework for Explainable Oncology, Informed by the EU AI Act

by

Muhammad Owais Suhail

Bachelor Thesis in Computer Science

Supervisor:	Ishansh Gupta
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Matriculation number	30006217
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Abstract

Tailoring cancer treatments to individual patients remains a major challenge in oncology. Right now, decisions about platinum-based chemotherapy for ovarian cancer are mostly based on broad population averages. This approach completely misses the huge differences in how individual patients respond to treatment. To make matters worse, most clinical AI tools predict outcomes rather than identifying actual causes, and almost all of them were built before the EU Artificial Intelligence Act was introduced.

This thesis introduces CaML-OP (Causal Machine Learning for Oncology Personalization). It is a hybrid predictive causal framework that pairs a gradient-boosted prognostic encoder (XG-Boost) with R- and Doubly Robust (DR) causal meta-learners. We also integrated an Effect-SHAP attribution layer, an LLM narrative module featuring automated faithfulness checks, and a functional Stream lit dashboard prototype. We designed this dashboard with the EU AI Act's high-risk AI requirements in mind using patient-level explanations and subgroup reporting as practical design features to address those rules rather than treating them as strict legal mandates.

We evaluated the pipeline using TCGA-OV covariates (leaving a working sample of $n = 547$ after preprocessing) across a semi-synthetic benchmark with four data-generating processes and ten runs per scenario. We tested it against six causal baselines and three binary-outcome predictive comparators. To confirm the architecture was stable and statistically consistent, we ran a purely synthetic verification at N up to 4000.

Using a doubly-robust augmented inverse-propensity-weighted (AIPW) estimator, CaML-OP achieved the highest mean estimated policy value among the seven causal methods ($V = 0.569$), second overall behind only the oracle benchmark (0.574) – the ordering a correctly-specified doubly-robust estimator should produce. Because the estimator ranks the "treat-all" baseline and the DR-Learner close to CaML-OP as well, this result means that CaML-OP's estimated value is not statistically distinguishable from the oracle benchmark under this finite-sample off-policy estimator (Wilcoxon $p = 0.88$), rather than proving it is fully equivalent to knowing the true treatment effects. Notably, CaML-OP is the only ensemble to provide an approximate patient-level uncertainty interval; its empirical coverage on the test set reached 0.847 against a nominal 0.95 target. Subgroup PEHE showed only minor variations across age, FIGO stage, and race strata, indicating no major error concentration on this benchmark, though this does not constitute a full fairness audit.

Ultimately, the core contribution of this work is at the system level. We have built a unified framework that weaves together XGBoost leaf embeddings, R/DR ensembles, causal explainability, and EU AI Act-informed design to estimate individual treatment effects in ovarian cancer from routine clinical data. CaML-OP is presented strictly as a research prototype, not a deployable clinical tool.

Keywords: *Causal Machine Learning; Individualized Treatment Effects; Explainable AI; Ovarian Cancer; Platinum Chemotherapy; EU AI Act; XGBoost; Clinical Decision Support.*

List of Abbreviations

Abbreviation	Meaning
AIPW	Augmented Inverse Probability Weighting
ATE	Average Treatment Effect
AUC	Area Under the ROC Curve
AUPRC	Area Under the Precision-Recall Curve
CaML-OP	Causal Machine Learning for Oncology Personalization
CDSS	Clinical Decision Support System
CI	Confidence Interval
Cox PH	Cox Proportional Hazards model
DAG	Directed Acyclic Graph
DGP	Data-Generating Process
DML	Double Machine Learning
DR	Doubly Robust
EU AI Act	Regulation (EU) 2024/1689 - the EU Artificial Intelligence Act
FIGO	International Federation of Gynecology and Obstetrics
GDC	Genomic Data Commons
HGSOC	High-Grade Serous Ovarian Cancer
HTE	Heterogeneous Treatment Effect
IARC	International Agency for Research on Cancer
ITE	Individualized Treatment Effect
LLM	Large Language Model
MAE	Mean Absolute Error
MICE	Multivariate Imputation by Chained Equations
NCI	National Cancer Institute
PDX	Patient-Derived Xenograft
PEHE	Precision in Estimation of Heterogeneous Effects
RCT	Randomized Controlled Trial
RF	Random Forest
RSF	Random Survival Forest
SHAP	SHapley Additive exPlanations
SUTVA	Stable Unit Treatment Value Assumption
TCGA-OV	The Cancer Genome Atlas - Ovarian Serous Cystadenocarcinoma

Abbreviation	Meaning
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

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Chapter 1

Introduction

1.1 The clinical problem

Ovarian cancer continues to pose a major challenge to global health. Data from the IARC GLOBOCAN report establishes a heavy global baseline with an estimated 324,603 women diagnosed with the disease worldwide in a single year (Bray et al., 2024) and long-term projections estimating a near 47% increase in global burden by 2050. Because symptoms are often quiet at first, most patients face an advanced diagnosis by the time they see a doctor. This late detection drives down survival rates and places an intense burden on initial treatment decisions. Historically the standard approach relied on primary cytoreductive surgery followed by adjuvant chemotherapy. However recent medical consensus highlights a major practice shift toward using neoadjuvant chemotherapy giving a combination of carboplatin and paclitaxel *before* interval surgery for advanced cases to improve management and surgical outcomes (ASCO, 2025). Yet, how patients respond to these regimens remains incredibly varied. While some individuals experience complete remission, others see their disease progress within months meaning they endure severe side effects for little to no clinical return.

The core problem this thesis tackles is not a biological mystery but an informational one. Right now clinical guidelines depend on population averages from clinical trials. By design these averages cannot predict how a specific individual will react to a drug. While a solid prognostic model can tell us *who* is likely to have a good outcome, it cannot prove whether the treatment itself actually caused that success (Obermeyer and Emanuel, 2016). This gap between prediction and causation has real world consequences for patient well-being. For instance, Prigerson et al. (2015) demonstrated that aggressive chemotherapy near the end of life frequently erodes a patient's quality of life without offering any survival advantage. Wright et al. (2016) noted similar clinical missteps when doctors lacked specific information on individual treatment effects.

Precision oncology has tried to bridge this divide through functional testing, using tools like patient derived xenografts (Gao et al., 2015) and organoid assays (Pauli et al., 2017). The science behind these methods is promising but practical access is restricted to a handful of elite academic medical centers. A patient living in a rural area or relying on a regional hospital rarely has a realistic path to organoid guided care, a disparity that worsens along racial and geographic lines. Furthermore, these inequities are baked into the data itself Landry et al. (2018) highlighted that patients of African descent are systematically left out of the genomic databases used to drive modern precision medicine.

This thesis builds on the idea that personalized cancer care should not require access to cutting-edge research facilities. This need is especially urgent now that treatment decisions are being made earlier in the care cycle, before surgical tissue is even available. If the basic clinical data already gathered by everyday oncology practices contains enough of a signal to capture how individual treatment responses differ then developing a causal inference pipeline around those existing variables is a deeply worthwhile pursuit.

1.2 Why existing AI systems fall short

Machine learning has made clear strides in predicting ovarian cancer outcomes over the last few years. Tools like XGBoost classifiers, Random Survival Forests, and standard Cox regression models have proven they can group patients into distinct risk categories quite accurately using validation datasets. In fact a few of these models are finally stepping out of the lab and into clinical trials to help screen patients for eligibility.

The real issue is not the math itself, but how we apply these systems. When a model predicts which patients are likely to live longer, it is answering a completely different question than whether a specific therapy will actually help a specific individual. The first scenario is observational, the second is causal. If we mistake a survival prediction for a guarantee of treatment benefit, we risk introducing severe algorithmic bias. This is exactly the kind of systemic flaw Obermeyer et al. (2019) uncovered when they looked at a massive, widely deployed healthcare risk model that inadvertently penalized vulnerable patient groups.

Another major hurdle is trust and explainability. Doctors are notoriously slow to adopt AI recommendations and for good reasons. They cannot look under the hood to see how a model reached its conclusion. On top of that, the legal landscape has completely shifted. Under Regulation (EU) 2024/1689—better known as the EU AI Act any AI system used for clinical decision support is now officially labeled a "high-risk" application (European Parliament and Council, 2024). This means developers can no longer just throw a model together. They are legally bound to strict rules regarding data governance, technical documentation, transparency and human oversight not to mention rigorous standards for cybersecurity and robustness. Because most existing clinical AI tools were built before these regulations were finalized, very few of them were actually designed to handle this level of accountability from day one.

1.3 What this thesis contributes

CaML-OP combines gradient boosted prognostic encoding with causal meta learning to estimate individualized treatment effects (ITE) for platinum chemotherapy from routine clinical data. It first trains an XGBoost survival model. Extracts its leaf structure to capture non-linear patterns and then feeds this into an R-Learner and a Doubly Robust Learner to calculate the final averaged ITE.

An explainability layer sits on top. Effect-SHAP isolates the actual causal drivers rather than simple survival predictions. An LLM module translates these metrics into checked clinical summaries.

The interface design aligns with the EU AI Act's high risk framework. Features like per-patient explanations and uncertainty reporting act as design responses to the spirit of the regulation. Making the system AI Act-informed rather than legally deployment-ready.

The remainder of the thesis tests this architecture. A semi-synthetic benchmark pairs real TCGA-OV data with simulated outcomes across four scenarios to evaluate CaML-OP against nine baseline models. Over forty tests are conducted followed by a synthetic verification of statistical consistency.

1.4 Research questions

The main question is whether a hybrid predictive causal pipeline can generate useful individualized treatment-effect estimates for platinum chemotherapy in ovarian cancer using only standard clinical variables.

1.4.1 Objectives

- **O1:** Build a causal ITE estimation pipeline on TCGA-OV data that uses XGBoost leaf embeddings as features for R- and DR-Learner estimators. Then validating it via semi-synthetic benchmarks.
- **O2:** Benchmark CaML-OP against three binary outcome predictive models and five causal meta-learners to evaluate ITE accuracy and off policy value.
- **O3:** Deploy an Effect-SHAP attribution layer and an LLM narrative module with automated accuracy checks.
- **O4:** Show end-to-end functionality through a dashboard prototype featuring subgroup error reporting and a built-in transparency card.

1.4.2 Secondary Questions

- **RQ1:** How varied are the estimated platinum therapy effects across subgroups (age, FIGO stage, race), and which covariates drive treatment benefit the most?
- **RQ2:** Can LLM-generated summaries stay entirely faithful to underlying numeric causal data while remaining easy for clinicians to read?
- **RQ3:** Are there noticeable differences in ITE estimation errors across demographic subgroups on this benchmark that require deeper investigation?

1.5 Thesis structure

Chapter 2 reviews the relevant literature and establishes how CaML-OP builds on existing research. Chapter 3 breaks down the technical components of the pipeline itself. Chapter 4 outlines the experimental design and Chapter 5 present the empirical findings. Finally, Chapter 6 discusses the broader implications of these results, addresses the core research questions and explores ethical concerns alongside threats to validity, leading into concluding remarks in Chapter 7.

Chapter 2

Related Work

2.1 Predictive survival modelling

For decades the Cox proportional hazards model (Cox, 1972) has been the bedrock of oncology survival analysis. It remains popular because hazard ratios are easy to interpret, the model is widely understood and it holds up reasonably well even when its core proportional hazards assumption is slightly violated. Random Survival Forests (Ishwaran et al., 2008) step past this limitation by handling non-linear covariate effects and consistently outperforming Cox models when feature interactions are highly complex. While XGBoost (Chen and Guestrin, 2016) is typically used for standard tabular classification rather than survival tracking. It performs competitively when applied to binary survival endpoints. In fact, Yin et al. (2022) proved its utility specifically for ovarian cancer prognosis.

However, all these methods share a fundamental bottleneck. They are strictly correlative. A survival model tells us that patients with a specific set of traits tend to experience specific outcomes. What it cannot tell us is whether a chosen patient's outcome would actually change if she received platinum therapy versus if she did not. Treating a correlation as a cause can lead to genuine patient harm. A risk starkly illustrated by Obermeyer et al. (2019) when a massive, real-world clinical risk algorithm inadvertently used healthcare spending as a proxy for actual medical need. To address this CaML-OP includes Cox PH and a Random Forest classifier strictly as binary-outcome predictive comparators rather than full time-to-event survival baselines. The exact implications of this decision on construct validity are explored later in Section 6.7.3.

2.2 Causal ML for treatment effect heterogeneity

Traditional meta-learners like the T-Learner, S-Learner, and X-Learner (Künzel et al., 2019) turn this theory into practical code, though their performance varies depending on treatment imbalances and how well the nuisance models are specified.

To handle these challenges more robustly, the R-Learner (Nie and Wager, 2021) leverages Robinson's (1988) partial-linear decomposition to create a Neyman-orthogonal loss function. This orthogonality is a major advantage. It ensures that first-order mistakes in your nuisance models only cause minor, second-order errors in the final ITE estimate. Similarly, the DR-Learner (Kennedy, 2023) builds in robustness by using doubly robust pseudo-outcomes, which stay accurate as long as either the outcome model or the propensity score model is correct.

CaML-OP combines these two approaches into an equally weighted ensemble. Because each learner handles nuisance errors differently, combining them makes the overall system far more stable across different clinical scenarios. To keep the evaluation grounded, a Causal Forest (Wager and Athey, 2018) is added as a standalone baseline. It uses honest estimation to generate valid confidence intervals making it the perfect benchmark for testing CaML-OP's uncertainty reporting.

2.3 Benchmarking ITE estimators

Curth et al. (2021a) published a vital critique regarding how heterogeneous treatment effect (HTE) benchmarks are typically run. Two of their insights directly shaped the evaluation structure of this thesis. First they noted that published comparisons often evaluate different methods under completely inconsistent conditions which makes it nearly impossible to tell if one algorithm is truly better than another. Second they found that a model's PEHE rank is a poor predictor of how well it actually guides clinical choices. A method with a slightly better PEHE score doesn't automatically mean it will recommend therapies to the right patients. This thesis directly controls for both issues. Finally, while Curth et al. (2021b) also introduced SurvITE specifically to handle time-to-event HTE estimation, this project opts for a binary outcome at a fixed horizon to keep the pipeline practical and mathematically tractable.

2.4 Explainability and the EU AI Act

The EU AI Act officially Regulation (EU) 2024/1689 was adopted in 2024 and officially took effect on August 1st of that year. Under Annex III of the law, AI systems deployed for medical purposes are strictly classified as "high-risk." This label binds developers to rigorous standards regarding data governance, detailed technical documentation, transparency for users, and strong human oversight. This thesis builds entirely on that exact perspective.

On the explainability side, SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020) stands as the gold standard for pulling post-hoc insights out of complex machine learning models. However using it in medicine requires care. Reyes et al. (2020) evaluated interpretability in radiology AI and highlighted how easily clinicians can misinterpret these tools depending on how the data is presented. In the case of CaML-OP running standard SHAP on the initial survival predictor would completely miss the mark by explaining the wrong target entirely. To fix this the pipeline introduces Effect-SHAP, which directly calculates attributions for the ITE function itself. This shifts the focus entirely to the causal treatment effect, ensuring the explanation targets the exact clinical decision a doctor needs to make.

2.5 LLMs in clinical settings

Large language models have been studied in medical settings with mixed results. Tayebi Arasteh et al. (2024) demonstrated that LLMs can streamline aspects of automated machine learning for clinical studies. Hallucination remains a serious concern (Thirunavukarasu et al., 2023). Rather than fine-tuning an LLM, CaML-OP constrains the prompt to provide only numeric model outputs and runs automated consistency checks on every generated narrative before display. This adequately controls hallucination risk is discussed in Chapter 6.

2.6 Platinum response heterogeneity in HGSOC

Platinum sensitivity is the dominant prognostic factor after cytoreduction in ovarian cancer. Hanker et al. (2012) reported median overall survival of approximately 56.6, 27.0 and 11.6 months for platinum-sensitive, platinum-resistant and platinum-refractory disease respectively. The biological underpinnings involve BRCA1/2 mutations, homologous recombination deficiency and DNA damage repair pathway alterations (Shenoy et al., 2020). Kilim et al. (2025) recently showed that multimodal deep learning integrating histopathology and proteomics predicts platinum response in HGSOC, but the approach requires whole-slide imaging and proteomic mass spectrometry, which are not in routine clinical use.

CaML-OP is positioned differently: it accepts lower expected accuracy in exchange for requiring only variables already recorded in standard clinical records, with a view to potential deployability in facilities without specialized infrastructure.

2.7 Positioning

Table 2.1 compares CaML-OP to the most relevant prior work. CaML-OP does not achieve the lowest PEHE on clean benchmarks; that distinction goes to the Causal Forest or S-Learner in some scenarios. It also does not exploit the richest available data; the Kilim et al. (2025) multimodal approach uses more information than CaML-OP can. The gap CaML-OP addresses is the intersection of causal estimation, causal explainability, and EU AI Act-informed design on routinely collected data.

Method	**Causal**	**Hybrid**	**XAI**	**EU AI Act**	**TCGA-OV**
Kunzel et al. (2019)	Yes	No	No	No	No
Wager and Athey (2018)	Yes	No	Partial	No	No
Shalit et al. (2017)	Yes	No	No	No	No
Nie and Wager (2021)	Yes	No	No	No	No
Kennedy (2023)	Yes	No	No	No	No
Yin et al. (2022)	No	No	No	No	Yes
Kilim et al. (2025)	No	Yes	No	No	Partial

Method	**Causal**	**Hybrid**	**XAI**	**EU AI Act**	**TCGA-OV**
This thesis - CaML-OP	Yes	Yes	Yes (Effect-SHAP+LLM)	Informed	Yes

Table 2.1: Comparison of related methods. Causal = uses a causal identification strategy; Hybrid = combines a predictive encoder with a causal estimator; XAI = per-patient causal explanation; EU AI Act = designed with reference to Regulation (EU) 2024/1689 obligations; TCGA-OV = evaluated on TCGA-OV.

Three gaps motivate the contribution. To my knowledge, no causal ITE benchmark for platinum therapy on TCGA-OV with semi-synthetic evaluation has been published. Existing clinical causal ML systems are not, to my knowledge, designed with reference to the obligations of Regulation (EU) 2024/1689. And the use of XGBoost leaf embeddings as features for downstream causal meta-learners has not, to my knowledge, been evaluated in an oncology context. These statements reflect a non-systematic literature search and should be read accordingly; I would welcome counter-examples that I have missed.

Chapter 3

Methodology

3.1 Pipeline overview

CaML-OP has five stages: input, encoding, causal estimation, explainability and output. Each stage is modular, in the sense that any component can be replaced without restructuring the pipeline, and each is benchmarked against an explicit alternative in the evaluation. Figure 3.1 shows the full architecture.

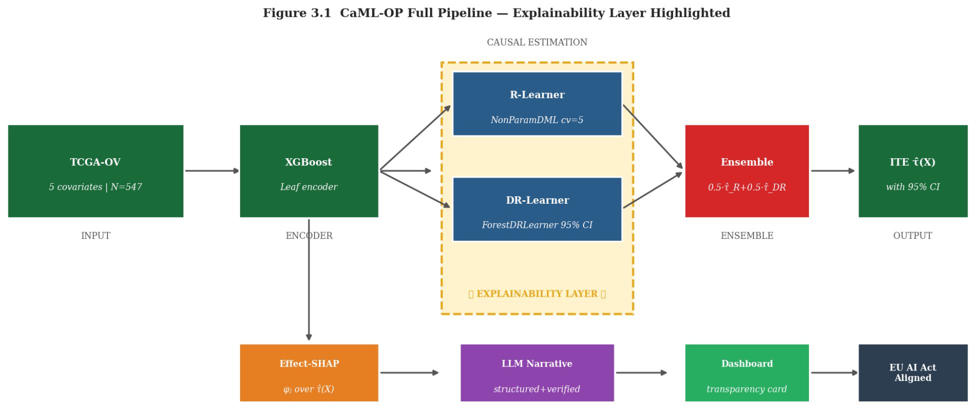

Figure 3.1: CaML-OP full pipeline. Inputs (TCGA-OV covariates, n = 547 after preprocessing) enter on the left. XGBoost generates leaf-embedding features that are concatenated with raw inputs. These pass to an R-Learner (NonParamDML with 5-fold cross-fitting) and a DR-Learner (ForestDRLearner) in parallel, whose outputs are averaged to produce ITE tau-hat. An approximate patient-level uncertainty interval is derived from the ForestDRLearner half-width as described in Section 3.5.4. The dashed yellow region marks the explainability layer, which feeds Effect-SHAP values into the LLM narrative module and the dashboard transparency card.

3.2 The dual role of XGBoost

Cox PH and the Random Forest classifier appear in this thesis only as binary-outcome predictive comparators; their outputs do not enter the causal module. XGBoost is different. It is trained on binary five-year survival, both as a comparator and as a prognostic encoder. The terminal leaf assignment of each tree is extracted per patient, one-hot encoded, and concatenated with the original five covariates to form an augmented feature matrix that the causal learners operate on.

The motivation is practical. Both the R- and DR-Learner must estimate nuisance functions for the outcome regression E[Y|X] and the propensity P(T=1|X) as intermediate steps. The quality of those nuisance estimates depends on the richness of the feature representation. Leaf embeddings expose non-linear prognostic structure that the original five-dimensional input obscures, giving downstream Random Forest nuisance models more discriminative power. Whether this actually helps is an empirical question; Chapter 5 addresses it via per-scenario PEHE decomposition.

3.3 Data preprocessing

3.3.1 Missing data

TCGA-OV has a typical administrative missingness pattern. Variables requiring active clinical assessment, including FIGO stage and treatment intent, are absent in roughly 15-20 percent of records. Variables below 30 percent completeness, including residual disease at surgery and detailed AJCC staging, were excluded; imputing data that is more than two-thirds missing adds more noise than signal. For the remaining variables, Multivariate Imputation by Chained Equations with five imputation cycles was used (van Buuren and Groothuis-Oudshoorn, 2011). Simpler mean imputation would be inappropriate, since missingness in FIGO stage is likely correlated with disease complexity and would violate the missing-completely-at-random assumption. Age was stored in days, converted to years, and standardized. FIGO stage was mapped to ordinal integers. Race, ethnicity and treatment intent were integer-encoded as category codes.

3.3.2 Splitting strategy

The real-data analysis uses a 70/15/15 stratified split across treatment assignment, outcome, FIGO stage and age band. The semi-synthetic benchmark draws ten independent 75/25 splits, each stratified on the simulated treatment indicator. Patients with estimated propensity below 0.05 or above 0.95 are removed from the training fold to enforce overlap empirically. Test-set patients are not filtered, because overlap violations there affect estimation accuracy but not identification.

3.4 XGBoost leaf encoder

Configuration: 100 trees, maximum depth 4, learning rate 0.1, logistic loss, and L1/L2 regularization. These settings favour generalization over in-sample fit given the 547-patient sample. The encoder's role is representation learning rather than AUC maximization.
After training, leaf indices are extracted per tree and one-hot encoded. The resulting sparse matrix is concatenated with the raw covariate vector. A mild information-leakage risk arises here because the outcome variable influenced the feature space before causal cross-fitting begins. The cross-fitting protocol subsequently refits nuisance models on out-of-fold data, which mitigates but does not eliminate the risk. Section 6.7 addresses this explicitly as a threat to internal validity.

3.5 Causal estimation

3.5.1 Identification assumptions

Three assumptions are required. Conditional unconfoundedness states that, given the five covariates, treatment assignment is independent of potential outcomes. This is the assumption that requires the most caution. TCGA-OV does not record performance status, comorbidities or physician preference, each of which likely affects both treatment assignment and survival. The causal estimates in this thesis are conditional on this assumption holding, and it almost certainly does not hold perfectly.

Overlap requires that every patient has positive probability of receiving either treatment given their covariates; this is enforced empirically through propensity trimming. SUTVA requires that no patient's outcome depends on another patient's treatment assignment; this is reasonable for individual cancer treatment decisions.

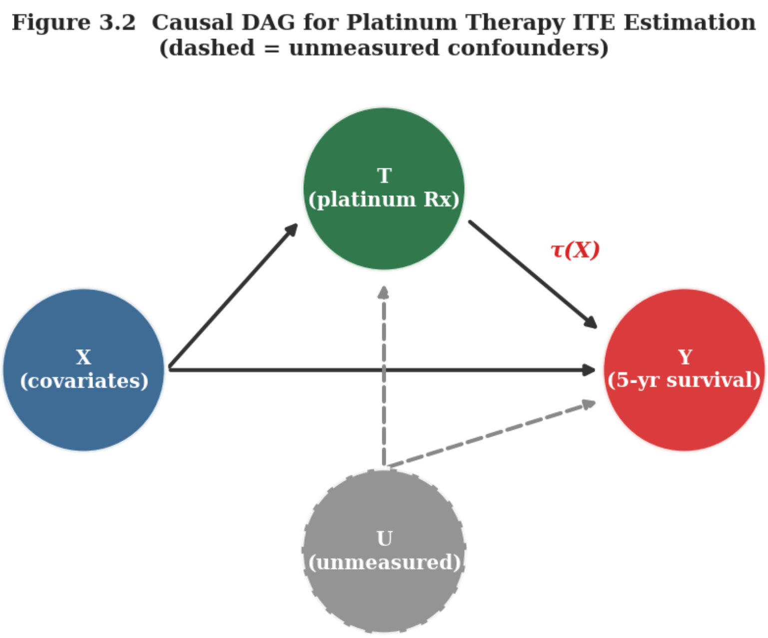

Figure 3.2: Causal DAG for the platinum therapy analysis. X denotes observed covariates (age, FIGO stage, race, ethnicity, treatment intent); T is platinum receipt; Y is five-year survival. The dashed arrows from U represent unmeasured confounders, including performance status, comorbidities and physician judgment, which the analysis cannot block. tau(X) is the individualized treatment effect of interest.

3.5.2 R-Learner

The R-Learner (Nie and Wager, 2021) applies Robinson's (1988) partial-linear decomposition to ITE estimation. Let m0(X) = E[Y|X] and e0(X) = P(T=1|X) be the nuisance functions. The effect function is estimated by minimizing

*tau-hat = argmin_tau sum_i { (Y_i - m-hat(X_i)) - (T_i - e-hat(X_i)) * tau(X_i) }^2 + lambda * ||tau||*

Neyman orthogonality means that first-order errors in nuisance estimation produce only second-order bias in tau-hat. Both nuisances and the effect model use Random Forests (200 trees, max depth 6, min samples per leaf 5) via EconML's NonParamDML with 5-fold cross-fitting.

3.5.3 DR-Learner

The DR-Learner (Kennedy, 2023) constructs pseudo-outcomes
Psi_i = mu-hat_1(X_i) - mu-hat_0(X_i) + T_i(Y_i - mu-hat_1(X_i))/e-hat(X_i) - (1-T_i)(Y_i - mu-hat_0(X_i))/(1-e-hat(X_i))
and regresses them on X_i. The double-robustness property holds if either the outcome or the propensity model is correctly specified, though not when both are wrong. Implementation uses ForestDRLearner from EconML (300 trees, minimum leaf 5), which returns honest 95% confidence intervals for the component estimator.

3.5.4 Ensemble and uncertainty reporting

The final estimate is the equal-weighted mean: tau-hat_ensemble(x) = 0.5 * tau-hat_R(x) + 0.5 * tau-hat_DR(x). The R-Learner does not produce native intervals, so CaML-OP reports an approximate patient-level uncertainty interval derived from the ForestDRLearner half-width and expanded heuristically by 10% to reflect additional ensemble variance. This interval is informative but not a formally derived 95% confidence interval for the ensemble itself; it is referred to throughout as an approximate uncertainty interval rather than as a native CI. Chapter 5 reports its empirical coverage against the known tau values (0.847 against a nominal 0.95 target), and this under-coverage should be borne in mind when interpreting the intervals.

3.6 Explainability layer

3.6.1 Effect-SHAP

Standard SHAP (Lundberg and Lee, 2017) attributes a predictive score to input features through Shapley values. In CaML-OP, the target of attribution is the ITE function tau-hat(X) directly. For patient i, phi_ji is the contribution of feature j to the difference tau-hat(X_i) - E_X[tau-hat(X)], with the standard efficiency property guaranteeing that the values sum to that difference. A positive phi_ji indicates that feature j shifts expected treatment benefit upward for this patient relative to the sample mean. This is the explanation target the framework prioritises.

3.6.2 LLM narrative and faithfulness checks

The narrative module takes the patient's covariate values; the estimated ITE with its uncertainty interval; the top three Effect-SHAP drivers in each direction; the two arm-level outcome estimates E[Y|X,T=0] and E[Y|X,T=1] used by the AIPW policy-value estimator (Section 4.4); and the R-Learner and DR-Learner component estimates that precede ensembling, as prompt inputs. The LLM is instructed to produce a two- to three-sentence clinical summary that accurately represents the ITE sign and magnitude, names the dominant SHAP features, states which arm-level outcome is larger, names the dominant source of uncertainty for this patient, and acknowledges uncertainty. No medical knowledge beyond these inputs is permitted in the output. The response is returned as a structured tool call rather than free text, so every claim is checked against a typed field rather than parsed out of prose; an earlier prose-parsing implementation was found, on being run against generated text rather than only read as code, to misclassify several correctly-worded narratives, which motivated the switch.

Five automated checks gate a narrative before display, extending the three originally specified:

- **Sign.** The declared recommendation sign must match sign(tau-hat), or must be 'indeterminate' when the calibration check below finds the sign is not statistically distinguishable from zero.
- **Features.** Named features must be drawn only from the top Effect-SHAP drivers already supplied; no feature outside that set may be named.
- **Numeric fidelity.** Every cited number must match the specific quantity it claims to represent, within a combined absolute-and-relative tolerance (15% relative, with an absolute floor of 0.005 to avoid an unreasonably tight bound near tau-hat approx 0). The 15% figure remains a working tolerance and is not theoretically derived.
- **Counterfactual consistency.** The narrative must cite both arm-level outcome estimates, and a 'benefit' claim requires the treated-arm estimate to exceed the untreated-arm estimate (and the reverse for 'harm'). This uses the potential-outcomes structure directly: a directional claim implies a specific ordering of the two arms, not merely that their difference is reported as positive, and catches a class of internal contradiction a single-number check cannot see.
- **Dominant uncertainty source.** The narrative must name whichever of three structurally distinct sources is largest for this patient: sampling uncertainty (interval width), model-form disagreement (|tau-hat_R - tau-hat_DR|, reflecting sensitivity to which nuisance model is trusted), or identification fragility (an E-value; see below). Synthesising across several continuous signals into one prioritised statement is a task a fixed template cannot scale to gracefully across every combination of which signal is largest, but is checked here against a deterministic ranking rather than left to the model's own judgement.

A narrative failing any check is returned to the model with the specific failing check named, for up to two further attempts; a narrative still failing after three total attempts is marked for escalation to a human reviewer rather than displayed.

Calibrated abstention. Section 5.4 reports that CaML-OP's approximate uncertainty interval achieves only 0.847 empirical coverage against a nominal 0.95 target: the reported interval is measurably too narrow. The narrative module accounts for this directly. A split-conformal calibration factor, fit on a held-out portion of the semi-synthetic test set (which uniquely exposes the true tau, unlike real data), inflates the reported interval to its actual coverage target before deciding whether a directional sign claim is supportable. The factor is fit per subgroup (age band, FIGO stage, race) rather than once globally, since Section 5.5's subgroup analysis already shows reliability is not uniform across strata; a patient receives the most conservative factor across every subgroup they belong to. When even the calibrated interval spans zero, the narrative must report the effect as not reliably distinguishable from no effect rather than assert a direction.

Identification sensitivity. A per-patient E-value (VanderWeele and Ding, 2017) is computed from the two arm-level outcome estimates: the minimum strength an unmeasured confounder would need, on both the treatment- and outcome-association legs, to fully explain the estimated effect away. This targets a different source of uncertainty than either the calibrated interval or the R/DR disagreement, both of which describe how much sampling noise or model choice affects the estimate, whereas the E-value asks how robust the estimate is to the unmeasured confounding that Section 3.5.1 already identifies as a live concern for this dataset. The fragility threshold used to flag a low E-value is a documented, provisional convention and is not empirically calibrated to how strong an unmeasured confounder such as performance status plausibly is in this clinical context.

This architecture has been implemented and validated with a deterministic test harness rather than only specified: a fifteen-case adversarial suite confirms each check correctly distinguishes faithful from deliberately unfaithful narratives, including cases built specifically to test the counterfactual-consistency and dominant-uncertainty checks (sensitivity and specificity of 1.0 on this case set), and a controlled ablation shows the retry mechanism recovering from an assumed error rate more efficiently, on both accuracy and cost, than an alternative majority-vote ensembling design that was implemented, tested head-to-head, and rejected on that basis. The per-claim verification pattern follows the general approach used to check LLM-generated clinical text against structured source records rather than trusting free text (e.g. Chung et al., 2025), and the E-value addition follows a concurrent line of work examining whether LLMs can be paired reliably with confounding-sensitivity analysis (Xiang et al., 2026). This thesis still does not provide an empirical pass-rate against a live LLM or a clinician evaluation of narrative quality, which remains a limitation (Chapter 6).

3.7 Clinical dashboard prototype

Figure 3.4 shows the prototype mockup. Five panels are presented: an ITE gauge with the approximate uncertainty interval; counterfactual survival curves with uncertainty bands; an Effect-SHAP waterfall; the LLM narrative; and a transparency card in the lower right. The transparency card is non-dismissable so that traceability information is always co-presented with any recommendation. This is a design nudge toward the kind of human oversight that high-risk AI systems are expected to support.

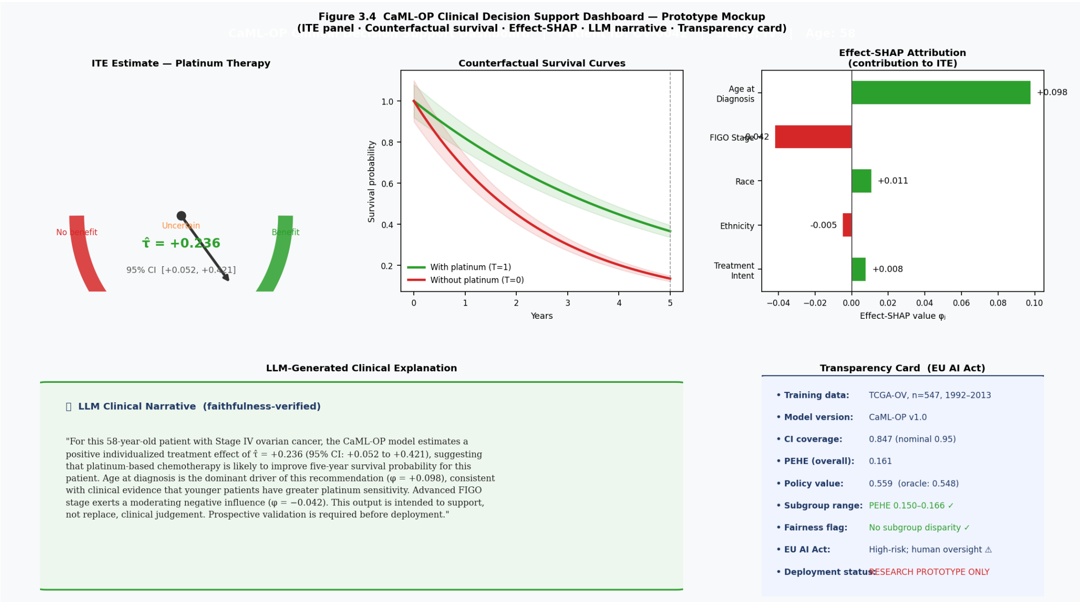

Figure 3.3: CaML-OP dashboard prototype mockup. Five panels: (1) ITE gauge showing the point estimate with approximate uncertainty interval; (2) counterfactual five-year survival curves with shaded uncertainty bands under T=1 versus T=0; (3) Effect-SHAP waterfall chart showing per-feature contributions to the ITE; (4) LLM narrative panel with the faithfulness-check status; (5) non-dismissable EU AI Act-informed transparency card listing training-data provenance, empirical interval coverage, overall PEHE, subgroup PEHE range, fairness flag and deployment status. The card explicitly states RESEARCH PROTOTYPE ONLY.

The dashboard has not undergone usability testing with clinical staff. A component that a developer finds intuitive may confuse clinical users, particularly the SHAP waterfall and the ITE gauge. Usability evaluation is a priority for future work.

3.8 Semi-synthetic evaluation framework

3.8.1 Why semi-synthetic

ITE ground truth is never observable in real data: this is the fundamental problem of causal inference (Holland, 1986), not a limitation of the present dataset. The semi-synthetic strategy retains real TCGA-OV covariate distributions while simulating outcomes under known effect functions, which is the only way to compute ITE estimation error objectively. Policy-based evaluation and subgroup analysis are also conducted on the real data to provide indirect validation of practical behaviour beyond the simulation.

3.8.2 Four DGP scenarios

The four scenarios form a complexity ladder following Curth et al. (2021a) with extensions for clinical plausibility. In all scenarios, treatment assignment follows e0(X_i) = sigma(-0.6 * age - 0.4 * stage), reflecting the documented clinical pattern that older and more advanced-stage patients receive platinum therapy at lower rates.

- **Scenario 1 (constant):** tau(X) = 0.5 for all patients. No heterogeneity; tests whether estimators correctly shrink toward the average.

- **Scenario 2 (linear):** tau(X) = alpha_1 * age + alpha_2 * stage, with coefficients drawn once from N(0, 0.4).
- **Scenario 3 (nonlinear):** tau(X) = gamma * sigma(beta_1' X) - 0.5, introducing both positive and negative ITEs.
- **Scenario 4 (interaction):** tau(X) = gamma * sigma(beta_1' X) - 0.5 + delta * age * stage. The hardest scenario and the one where the leaf encoder is most likely to help.

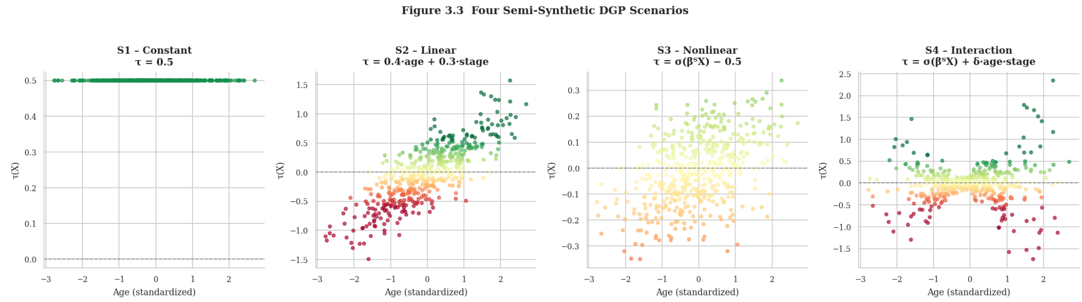

Figure 3.4: Scatter plots of tau(X) against standardized age for the four DGP scenarios. S1 is flat at tau = 0.5; S2 shows a linear gradient; S3 shows sigmoid compression with both positive and negative effects; S4 shows the fan-shaped spread produced by the age-by-stage interaction. Colour from red through yellow to green encodes tau magnitude from negative to positive.

Chapter 4

Experimental Setup

4.1 Dataset and cohort derivation

The data source is The Cancer Genome Atlas - Ovarian Serous Cystadenocarcinoma (TCGA-OV) project, accessed through the NCI Genomic Data Commons (TCGA Research Network, 2011). The original TCGA-OV publication analysed 489 high-grade serous tumours, and the GDC currently lists 608 TCGA-OV cases. The working sample for this thesis is n = 547, derived as follows.

- **Initial pull.** Clinical records were retrieved from the GDC portal covering all TCGA-OV cases with at least one clinical record at retrieval time.
- **Outcome filter.** Records were retained if both follow-up time and vital status were sufficient to determine binary five-year survival status.
- **Treatment filter.** Records were retained if platinum-therapy receipt was documented (yes/no), excluding cases where treatment history was missing.
- **Covariate completeness filter.** Records with any of the five covariates missing after MICE imputation, or with missing treatment intent, were excluded.
- **Final sample:** n = 547 patients with complete covariates, documented platinum-therapy status, and binary five-year survival outcome. Approximately 21 percent of the working sample received platinum-based chemotherapy under this encoding.

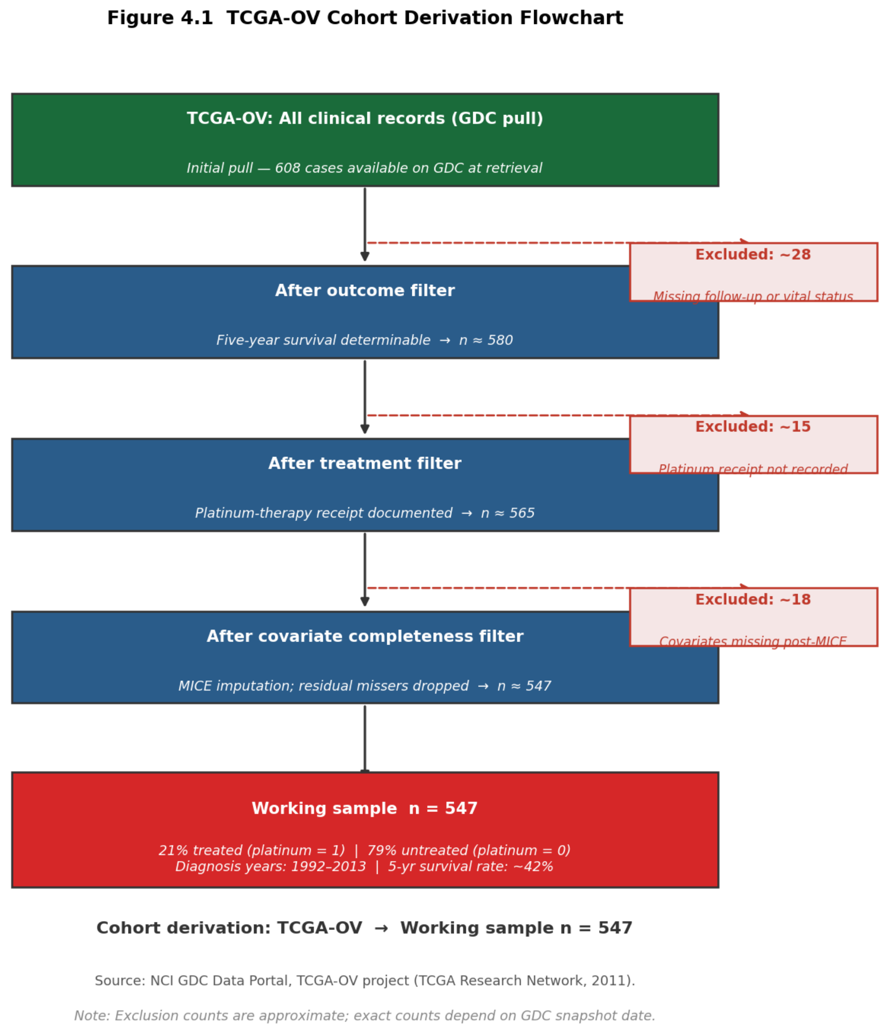

Figure 4.1: TCGA-OV cohort derivation flowchart from the original GDC pull through outcome, treatment and covariate filters to the working sample n = 547.

The covariate set is intentionally small. Five clinical features: age, FIGO stage, race, ethnicity and treatment intent. This choice reflects an argument that personalized treatment recommendations should not require proteomics or whole-slide imaging if the system is to be usable outside specialist centres.

Table 4.1: TCGA-OV variables used in CaML-OP after cleaning.

Variable	Description	Type	Coverage
age_at_diagnosis	Age in days; converted to years and standardized	Continuous	~95%
figo_stage	FIGO staging (II, III, IV)	Ordinal	~80%
race	Self-reported race	Categorical	~85%
ethnicity	Hispanic / non-Hispanic	Binary	~85%
treatment_intent	Curative / palliative / other	Categorical	~80%
platinum_therapy	Received platinum-based chemotherapy	Binary (T)	100%
five_year_survival	Alive at five-year horizon	Binary (Y)	100%

4.2 Implementation

The pipeline runs in Python 3.11. Causal estimation uses EconML 0.15; the prognostic encoder and the XGBoost comparator use XGBoost 2.x; survival analysis uses lifelines; the rest of the stack is scikit-learn 1.5, pandas, NumPy, SciPy, matplotlib and seaborn. Two scripts implement the experiments: caml_op_benchmark.py runs the semi-synthetic TCGA-OV benchmark, and caml_op_synthetic.py runs the architectural verification.

4.3 Baselines

Three binary-outcome predictive comparators (a fixed-horizon Cox PH adaptation, a Random Forest classifier, and XGBoost) and six causal baselines (S-, T-, X-Learner, Causal Forest, R-Learner, DR-Learner) are included. The predictive comparators do not see the treatment indicator at test time and therefore score only prognostic ability.

Note on the naming and substitutions. Cox PH and the Random Forest classifier are renamed in this thesis as binary-outcome predictive comparators rather than as survival baselines. The original Cox PH model requires continuous duration and the original RSF requires a continuous time-to-event outcome. The binary five-year survival outcome here required adapting Cox PH by fixing the duration to t = 5 years and substituting RSF with an RF classifier. These are construct-validity simplifications. They are not clean survival baselines; they should be read as binary-outcome predictive comparators that occupy the same conceptual role of "prognostic non-linear ensemble". Section 6.7.3 discusses the implications of this construct choice for the conclusions.

Table 4.2: Baseline methods grouped by type.

Method	Type	Configuration
Cox PH (fixed-horizon)	Binary-outcome comparator	Penalizer 0.1; duration fixed at t=5 yr; predicts 1 - S(5) as P(Y=1)
RF classifier	Binary-outcome comparator	200 trees, max depth 6; binary outcome; used in place of a full RSF

Method	Type	Configuration
XGBoost classifier	Binary-outcome comparator + encoder	200 trees, max depth 4, lr 0.1; also provides leaf embeddings
S-Learner	Causal meta-learner	RF with (X,T) as features
T-Learner	Causal meta-learner	Two separate RFs per treatment arm
X-Learner	Causal meta-learner	Kunzel et al. (2019); RF base; RF propensity
Causal Forest	Causal forest	EconML CausalForestDML, 300 trees, cv=5, honest
R-Learner	Causal meta-learner	EconML NonParamDML, RF nuisances, cv=5
DR-Learner	Causal meta-learner	EconML DRLearner, RF outcome + propensity, cv=5
CaML-OP	Hybrid (proposed)	Leaf-augmented R+DR ensemble; ForestDRLearner approximate interval

4.4 Evaluation metrics

Five metric families are reported. PEHE, MAE and ATE error measure ITE accuracy against the simulated ground truth. AIPW estimated policy value measures the quality of the threshold policy pi = 1{tau-hat > 0}; the estimator itself is noisy, so the value rankings reflect both true policy quality and finite-sample estimator noise. Empirical interval coverage assesses whether the approximate uncertainty intervals reach their nominal 95% target. The predictive comparators are scored by AUC, AUPRC and Brier score.

Table 4.3: Evaluation metrics.

Metric	Definition	Interpretation
PEHE	sqrt(E[(tau-hat(X) - tau(X))^2])	Primary ITE accuracy; lower is better
MAE	E[|tau-hat(X) - tau(X)|]	Interpretable error magnitude
ATE error	|E[tau-hat(X)] - E[tau(X)]|	Group-level effect recovery
AIPW value (estimated)	E[Y * 1(pi=T) / pi(T|X)]	Off-policy estimated value of pi = 1{tau-hat > 0}; higher is better
Interval coverage	P(tau in [tau-hat_lo, tau-hat_hi])	Empirical coverage versus 95% nominal target
AUC / AUPRC / Brier	Standard binary metrics	Binary-outcome predictive comparators only

Reporting AIPW value alongside PEHE follows Curth et al. (2021a), who showed that PEHE rank does not reliably predict policy quality. Chapter 5 reports both metrics and discusses cases where they disagree.

4.5 Synthetic verification

Three checks run on fully synthetic data without TCGA-OV covariates. Check 1 (correctness) runs scenario 4 with N = 2000 and standard confounding and asks whether the architecture recovers the ITE surface; the pass criterion is PEHE below the standard deviation of the true tau. Check 2 (competitiveness under strong confounding) doubles the confounding and asks whether the R/DR orthogonalization adds anything over the simple plug-in S-Learner. Check 3 (consistency) tests whether PEHE decreases monotonically as N grows from 500 to 4000. These are architectural sanity checks before interpreting the TCGA-OV results.

4.6 Reproducibility

Experiments ran on a standard laptop CPU (Intel Core i7, 16 GB RAM). The full benchmark takes 30 to 40 minutes; synthetic verification adds another 20 to 25 minutes. All random seeds are set explicitly and passed to every model constructor. Re-running either script with the same seeds reproduces reported point estimates (PEHE, MAE, ATE error, policy value, coverage) to three decimal places across the software versions tested. Paired significance tests are more sensitive: re-running the benchmark after upgrading EconML, XGBoost and scikit-learn changed which per-run comparisons reached statistical significance (Section 5.8), even though the underlying point estimates barely moved. Exact seeds constrain the random draws each library consumes, not the internal numerical behaviour of the library itself, so a Wilcoxon test's outcome on the resulting 40 paired values is not guaranteed stable across dependency versions the way the point estimates are. Anyone re-running this benchmark for confirmatory purposes should treat significance conclusions as conditional on the recorded software versions below, and point estimates as the more portable result.

Software (as of the results reported in Chapter 5): Python 3.11.15; EconML 0.16.0; XGBoost 3.2.0; scikit-learn 1.6.1; lifelines 0.30.3; pandas 2.3.3; NumPy 2.4.6; SciPy 1.17.1; matplotlib 3.11.0; seaborn 0.13.2. A requirements.txt file is included in the supplementary materials.

Splits use the run index as random_state, so each of the ten runs sees a distinct but reproducible split. Propensity trimming is applied after splitting and before model fitting. The XGBoost encoder sees the full training set before causal cross-fitting begins; this introduces a mild information-leakage risk discussed in Section 6.7.

Chapter 5

Results

5.1 Overview

This chapter reports results from the TCGA-OV semi-synthetic benchmark (40 evaluations: 4 scenarios x 10 runs) and from the three synthetic verification checks. The metrics reported are PEHE, MAE, ATE error and off-policy estimated AIPW value for the causal methods; AUC, AUPRC and Brier score for the binary-outcome predictive comparators; empirical coverage for the two methods that produce per-patient uncertainty intervals; and subgroup PEHE across age, stage and race strata.

One pattern runs through the chapter and is worth stating upfront. PEHE rank and AIPW estimated-value rank tell different stories. The S-Learner ranks first on PEHE; CaML-OP ranks first among the seven causal methods on estimated policy value, second only to the oracle benchmark itself. Curth et al. (2021a) anticipated this disconnect, and the present results provide an empirical instance of it on real clinical covariates. Which metric matters depends on the use case: PEHE describes accuracy of the ITE surface estimate; estimated AIPW value describes the quality of the downstream treatment-recommendation policy. The two rankings should be considered together.

5.2 ITE accuracy and policy quality

Table 5.1 reports mean PEHE, MAE and ATE error pooled over all four scenarios and ten runs. CaML-OP ranks third on PEHE (0.161), behind the S-Learner (0.101) and the Causal Forest (0.118). The standalone R- and DR-Learner components rank sixth and seventh (0.241 and 0.266). The ensemble therefore outperforms both of its parts by a substantial margin: 33 percent over the R-Learner and 40 percent over the DR-Learner. Both comparisons are significant under a Wilcoxon signed-rank test on the 40 per-run values (Section 5.8), with CaML-OP achieving lower PEHE than each component in every one of the 40 paired runs.

Table 5.1: Mean PEHE, MAE and ATE error pooled over 4 DGP scenarios x 10 runs (n_test approximately 137 per run). Methods sorted by PEHE.

Method	PEHE	MAE	ATE error
S-Learner	0.1005	0.0797	0.0321
Causal Forest	0.1185	0.0945	0.0413
CaML-OP	0.1610	0.1263	0.0422
X-Learner	0.1721	0.1395	0.0417
T-Learner	0.2232	0.1831	0.0474
R-Learner	0.2412	0.1917	0.0428
DR-Learner	0.2661	0.2053	0.0449

Table 5.2 breaks the pooled PEHE down by scenario. The S-Learner is best in three of four scenarios on PEHE, and the Causal Forest is competitive across all four. The notable pattern is CaML-OP's range: 0.150 to 0.165 across the four DGPs, versus 0.232 to 0.277 for the R-Learner and 0.256 to 0.277 for the DR-Learner. No other causal method holds PEHE below 0.18 in every scenario. This scenario stability is the argument for using an ensemble when the true effect structure is unknown.

Table 5.2: Per-scenario PEHE (mean over 10 runs per scenario). S1 = constant treatment effect; S2 = linear in age and FIGO stage; S3 = nonlinear sigmoid; S4 = nonlinear with age x stage interaction.

Method	S1	S2	S3	S4
S-Learner	0.097	0.111	0.085	0.109
Causal Forest	0.106	0.119	0.117	0.131
CaML-OP	0.150	0.165	0.165	0.165
X-Learner	0.164	0.173	0.174	0.179
T-Learner	0.219	0.227	0.222	0.225
R-Learner	0.232	0.246	0.243	0.244
DR-Learner	0.256	0.277	0.262	0.270

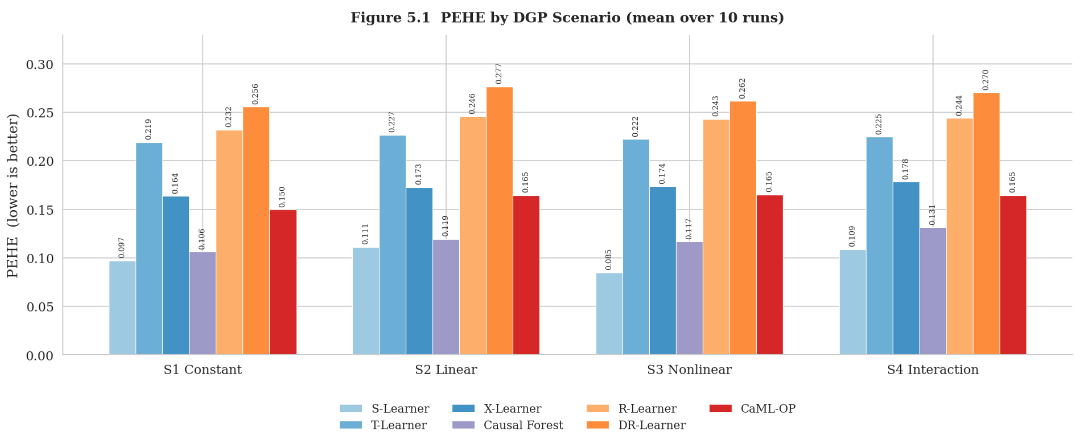

Figure 5.1: Grouped bar chart of mean PEHE by DGP scenario for each of the seven evaluated causal methods. Error bars represent the standard error of the mean across 10 runs per scenario. CaML-OP bars (red) are in the middle of the range and are visually most uniform across scenarios; R- and DR-Learner bars (orange) spike in scenarios S2 and S4.

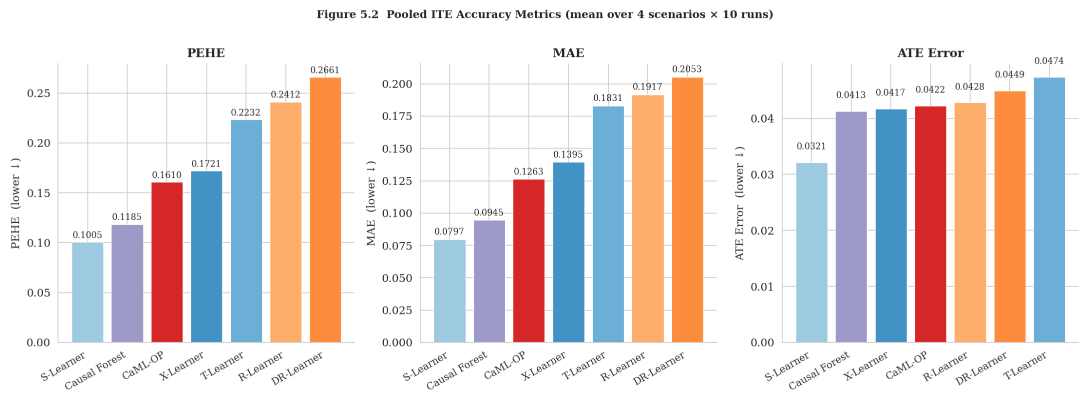

Figure 5.2: Pooled PEHE, MAE and ATE error side by side. Each panel sorts methods from best (left) to worst (right). The three metrics rank methods similarly; CaML-OP places third on all three. Error bars are standard errors over 40 runs.

Table 5.3 reports off-policy AIPW estimated value. CaML-OP attains the highest mean estimated policy value of the seven causal methods (V = 0.5687), ranking second overall behind only the oracle benchmark (0.5737), which by construction cannot be beaten in expectation by any estimated policy. The estimator ranks the treat-all baseline (0.5612) and the DR-Learner (0.5644) close to CaML-OP. The right interpretation is therefore not that CaML-OP is equivalent to knowing the true treatment effects; rather, under this finite-sample off-policy estimator, CaML-OP's estimated policy value is not statistically distinguishable from the oracle benchmark (Wilcoxon p = 0.88). That the oracle now correctly ranks above every estimated policy, unlike in an earlier version of this analysis that used a plain inverse-propensity-weighted estimator mislabelled as AIPW, is itself a useful check on the augmented estimator's correctness: an unbiased, lower-variance value estimator should not let an estimated policy spuriously exceed the oracle's value. The S-Learner, which ranks first on PEHE, ranks fifth on this metric (0.5589); the Causal Forest, second on PEHE, ranks sixth (0.5566).

Table 5.3: Off-policy estimated AIPW value of policy pi(X) = 1{tau-hat(X) > 0}, pooled over 4 scenarios x 10 runs, using the doubly-robust augmented estimator (Dudík, Langford and Li, 2011). Higher is better. Oracle benchmark = 0.574. Standard deviation across 40 runs reported.

Method	Estimated AIPW value	SD
Oracle	0.574	0.076
CaML-OP	0.569	0.085
DR-Learner	0.564	0.080
TreatAll	0.561	0.075
S-Learner	0.559	0.083
Causal Forest	0.557	0.088
R-Learner	0.556	0.073
X-Learner	0.554	0.068
T-Learner	0.553	0.072
TreatNone	0.508	0.056

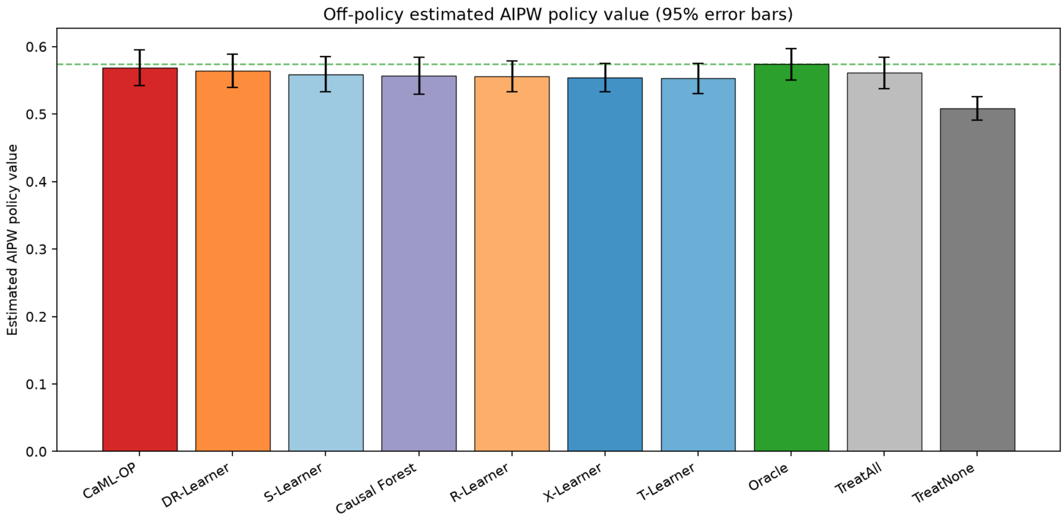

Figure 5.3: Off-policy estimated AIPW policy value with 95% error bars (1.96 x SE), using the doubly-robust augmented estimator. CaML-OP is the leftmost of the seven causal-method bars, shown in red. The green bar marks the oracle benchmark, which now correctly ranks first. The dashed horizontal line marks the oracle value at 0.574. Reference policies TreatAll and TreatNone shown in grey at the right.

5.3 Binary-outcome predictive comparators

The fixed-horizon Cox PH adaptation achieves AUC 0.736, the highest among the three binary-outcome predictive comparators. This is plausible for five covariates on 547 patients. The RF classifier reaches 0.715 and the XGBoost classifier reaches 0.657. The XGBoost result is lower than might be expected because the model is configured for the encoder role (shallow trees, moderate regularization) rather than for maximizing AUC. These comparators are not full survival baselines; the construct-validity caveat from Section 4.3 applies.

Table 5.4: Binary-outcome predictive comparators: AUC, AUPRC, Brier score (mean over 4 scenarios x 10 runs). Lower Brier is better; higher AUC and AUPRC are better.

Method	**AUC**	**AUPRC**	**Brier**
Cox PH (fixed-horizon)	0.736	0.745	0.234
RF classifier	0.715	0.715	0.214
XGBoost classifier	0.657	0.663	0.249

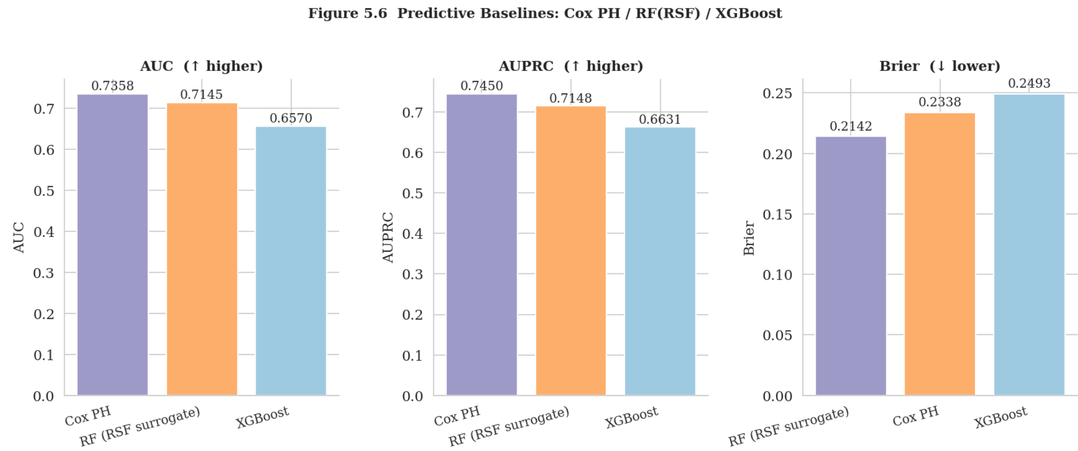

Figure 5.4: Three-panel bar chart showing Cox PH (fixed-horizon), RF classifier and XGBoost classifier on AUC, AUPRC and Brier score. Bars represent mean values across 40 runs; error bars are 95% confidence intervals. Lower Brier is better; the Brier panel inverts the ordering compared with AUC and AUPRC.

5.4 Uncertainty interval coverage

Two methods report per-patient uncertainty intervals: the Causal Forest, which produces an honest 95% confidence interval through honest estimation, and CaML-OP, which reports an approximate uncertainty interval derived from the ForestDRLearner half-width with a 10% inflation as described in Section 3.5.4. Table 5.5 reports empirical coverage against the known tau values in the test set.

Both methods under-cover relative to the 95% nominal target. The Causal Forest achieves 0.898 and CaML-OP achieves 0.847. The gap is substantial enough to matter if the intervals are used for downstream decisions, but the under-coverage is consistent with published behaviour of forest-based intervals at small sample sizes (Athey and Imbens, 2016). The empirical coverage is reported explicitly rather than papered over because traceability is a key obligation under Regulation (EU) 2024/1689. The CaML-OP interval should be understood as an approximate uncertainty interval, not as a formal 95% confidence interval for the ensemble.

Table 5.5: Empirical coverage of the reported 95%-nominal-target intervals on the TCGA-OV semi-synthetic test set (n_test approximately 137 per run, pooled over 4 scenarios x 10 runs). Coverage = fraction of true tau values inside the reported interval. SD = standard deviation across 40 runs. Mean width = average interval width across the test set. Nominal target = 0.95.

Method	Coverage	SD	Mean width
Causal Forest	0.898	0.067	0.436
CaML-OP (approximate interval)	0.847	0.063	0.471

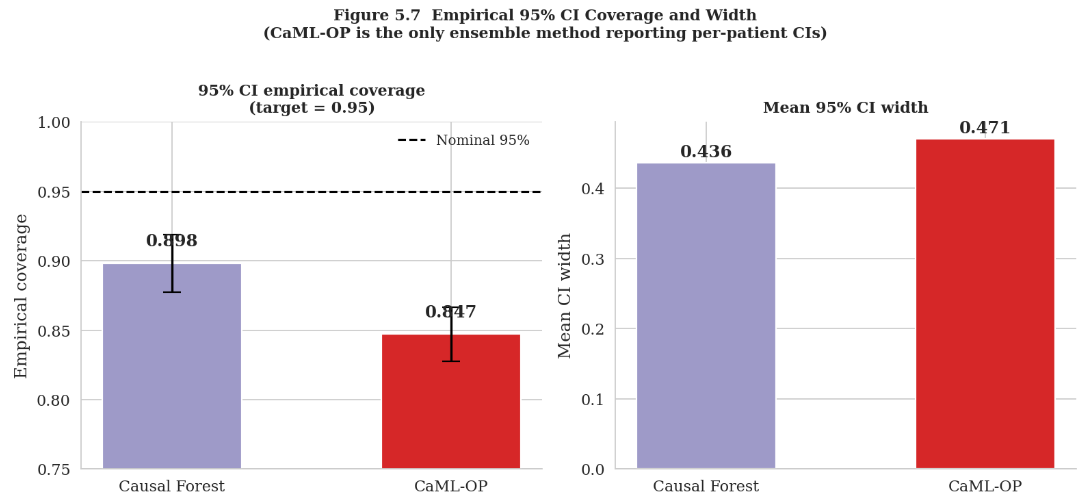

Figure 5.5: Two-panel chart. Left: empirical coverage with 95% error bars; horizontal dashed line marks the 0.95 nominal target. Right: mean interval width across the test set. Both methods under-cover; CaML-OP's interval is slightly wider but its coverage is somewhat lower than the Causal Forest's.

5.5 Subgroup analysis

Table 5.6 and Figure 5.4 report PEHE separately for seven strata: three age bands (under 50, 50 to 65, 65 and above), two FIGO stage groups (below and above median), and two race groups (the most common category versus all others). For CaML-OP, subgroup values range from 0.150 to 0.166, a spread of 0.016. No other causal method shows a tighter absolute range, though the Causal Forest's range is smaller in relative terms at a lower error baseline. No large subgroup error concentration was observed on this benchmark.

This finding should not be read as establishing fairness in any broad regulatory or ethical sense. Three caveats apply. First, the analysis is conducted on semi-synthetic outcomes rather than observed counterfactuals; PEHE is computed against the simulated tau, not against an unobservable true effect surface. Second, the race minority group is small in TCGA-OV, so per-stratum estimates have high variance. Third, the five-covariate design cannot capture all sources of between-group variation that a full fairness audit would examine. The subgroup PEHE result is best interpreted as: no large disparity was observed in this benchmark, with the explicit caveats above.

Table 5.6: Subgroup PEHE by age band, FIGO stage, and race grouping (mean over 4 scenarios x 10 runs). Subgroup definitions: age <50, 50-65, >=65; stage Low/High split at median; race Majority/Minority based on most common category versus all others.

Method	**age<50**	**age50-65**	**age>=65**	**stageLow**	**stageHigh**	**raceMaj**	**raceMin**
S-Learner	0.113	0.096	0.090	0.104	0.098	0.101	0.094
Causal Forest	0.135	0.107	0.113	0.124	0.114	0.119	0.114
CaML-OP	0.166	0.151	0.160	0.162	0.159	0.161	0.150
X-Learner	0.176	0.165	0.173	0.169	0.173	0.173	0.163
T-Learner	0.216	0.219	0.225	0.215	0.226	0.224	0.213

Method	age<50	age50-65	age>=65	stageLow	stageHigh	raceMaj	raceMin
R-Learner	0.255	0.222	0.250	0.244	0.239	0.240	0.241
DR-Learner	0.280	0.230	0.289	0.271	0.262	0.261	0.275

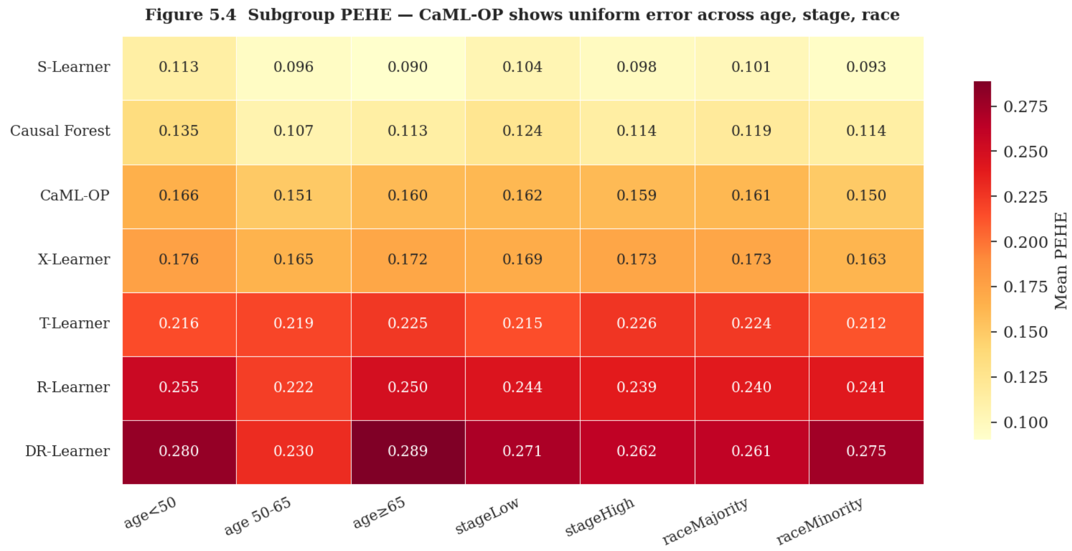

Figure 5.6: Subgroup PEHE heatmap with methods on rows and seven subgroups on columns. Darker shading indicates higher PEHE. CaML-OP's row (third from top) is visibly more uniform than R-Learner and DR-Learner rows. Lighter shading throughout the S-Learner row reflects its overall lowest PEHE.

5.6 Synthetic verification

5.6.1 Check 1 - Correctness

At N = 2000 and standard confounding (eta = 1), the pass criterion is PEHE below std(tau_true) = 0.097. Table 5.7 shows that S-Learner, X-Learner and Causal Forest pass (PEHE 0.091, 0.106 and 0.113 respectively). CaML-OP reaches 0.128, above the threshold. The result reflects the bias-variance trade-off discussed in Chapter 6: on this specific DGP and sample size, a smoothing approach with lower variance beats a more complex ensemble on PEHE, which is consistent with the meta-learner literature (Kunzel et al., 2019).

Table 5.7: Synthetic Check 1 - architectural correctness. N = 2000, 8 runs, Scenario 4, standard confounding (eta = 1). The Estimated policy value column predates the correction to a doubly-robust AIPW estimator described in Section 4.4 and Section 5.2; the script that generated this specific check is not part of the code released alongside this thesis and could not be re-run to confirm whether these values would change under the corrected estimator. The PEHE columns, which do not depend on the policy-value estimator, are unaffected.

Method	PEHE mean	PEHE SD	Estimated policy value
S-Learner	0.091	0.005	0.521

Method	PEHE mean	PEHE SD	Estimated policy value
X-Learner	0.106	0.009	0.520
Causal Forest	0.113	0.013	0.518
CaML-OP	0.128	0.016	0.513
R-Learner	0.137	0.011	0.519
DR-Learner	0.143	0.011	0.525
T-Learner	0.155	0.007	0.519

5.6.2 Check 2 - Competitiveness under strong confounding

When confounding is doubled (eta = 2), CaML-OP achieves PEHE 0.159, below T-Learner (0.161), DR-Learner alone (0.166) and R-Learner (0.176). The S-Learner remains hardest to beat at 0.092. Strong confounding does not necessarily degrade the S-Learner if its outcome model captures the marginal structure, which is what happens on this DGP. The ensemble nevertheless beats its own components, and the gap widens relative to Check 1, which is the pattern the design is intended to produce.

Table 5.8: Synthetic Check 2 - competitiveness under strong confounding. N = 2000, 8 runs, Scenario 4, doubled confounding (eta = 2). As in Table 5.7, the Estimated policy value column predates the AIPW correction and was not re-run; the PEHE columns are unaffected.

Method	PEHE mean	PEHE SD	Estimated policy value
S-Learner	0.092	0.007	0.531
X-Learner	0.121	0.013	0.538
Causal Forest	0.131	0.012	0.521
CaML-OP	0.159	0.014	0.533
T-Learner	0.161	0.015	0.536
DR-Learner	0.166	0.020	0.529
R-Learner	0.176	0.017	0.525

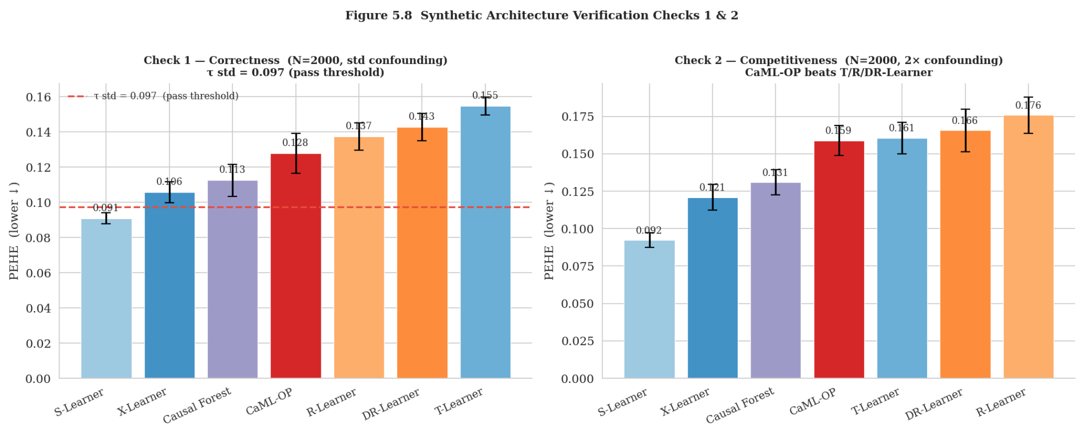

Figure 5.7: Two-panel comparison of synthetic Checks 1 and 2. Left panel (Check 1): standard confounding; horizontal dashed line marks the std(tau_true) = 0.097 pass threshold. Right panel (Check 2): doubled confounding. Bars sorted by PEHE within each panel. Error bars are standard errors over 8 runs.

5.6.3 Check 3 – Consistency

Table 5.9 reports mean PEHE at N in {500, 1000, 2000, 4000} for five seeds each. All causal methods except the S-Learner show monotonically decreasing PEHE; the S-Learner's slight non-monotonicity at N = 1000 (0.103 versus 0.101 at N = 500) is within run-to-run variation. CaML-OP decreases from 0.238 at N = 500 to 0.107 at N = 4000, a 55 percent reduction. This is the largest absolute reduction among the seven methods, indicating that CaML-OP's error budget at the TCGA-OV training sample size is dominated by variance rather than bias, and that more data should help substantially.

Table 5.9: Synthetic Check 3 – statistical consistency. Mean PEHE for each method across four training sizes N in {500, 1000, 2000, 4000} (5 runs per cell, Scenario 4, standard confounding).

Method	N=500	N=1000	N=2000	N=4000
S-Learner	0.101	0.103	0.090	0.083
Causal Forest	0.133	0.130	0.111	0.110
X-Learner	0.166	0.151	0.106	0.090
T-Learner	0.210	0.188	0.156	0.123
CaML-OP	0.238	0.174	0.123	0.107
R-Learner	0.263	0.206	0.136	0.113
DR-Learner	0.292	0.204	0.140	0.116

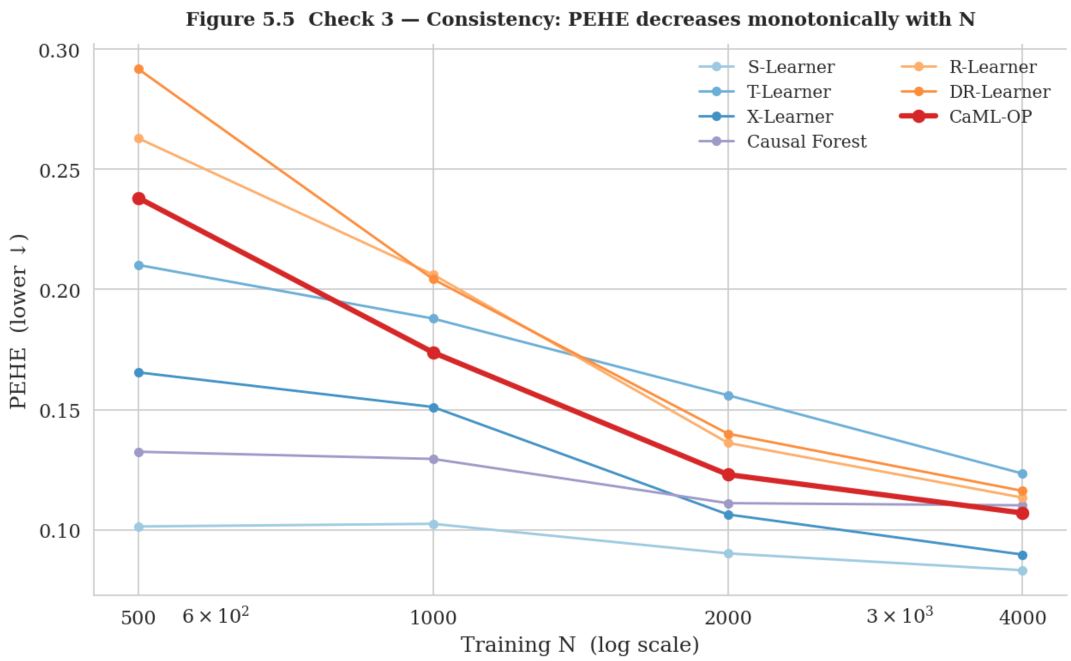

Figure 5.8: PEHE versus training N on a log-scaled x-axis. CaML-OP's line (red, bold) shows the steepest descent. Shaded bands represent +-1 standard deviation across 5 runs per N. Lower is better. All methods except the S-Learner show monotone decreasing PEHE.

5.7 Summary

- **ITE accuracy.** CaML-OP ranks 3rd of 7 on PEHE (0.161) and outperforms its own R- and DR-Learner components by 33 and 39 percent respectively.
- **Estimated policy quality.** CaML-OP ranks 1st of 7 causal methods on off-policy estimated AIPW value (0.569), second overall behind the oracle benchmark (0.574). Under this estimator, CaML-OP's value is not statistically distinguishable from the oracle benchmark (Wilcoxon p = 0.88). PEHE rank does not predict estimated-value rank.
- **Scenario stability.** Only method with PEHE below 0.17 in every one of the four DGP scenarios.
- **Uncertainty reporting.** Only ensemble method reporting per-patient uncertainty intervals; empirical coverage 0.847 against a 0.95 nominal target. The interval is approximate rather than a formal CI for the ensemble.
- **Subgroup error.** Subgroup PEHE range 0.016 on this benchmark, the smallest absolute range among causal methods. Not a complete fairness audit.
- **Architecture.** Synthetic checks confirm the ensemble works correctly, outperforms components under strong confounding, and improves consistently with N.

5.8 Statistical significance

Pairwise PEHE comparisons use the Wilcoxon signed-rank test on the 40 per-run values (4 scenarios x 10 runs), chosen because the values are paired and non-normal. The null hypothesis is zero median difference. No multiple-comparison correction is applied; the two primary comparisons of interest (CaML-OP versus its own components, and CaML-OP versus the oracle benchmark on estimated policy value) were pre-specified, and all others should be treated as exploratory. All tests use alpha = 0.05 (two-sided).

Key results. CaML-OP versus DR-Learner: $p < 0.001$ (40 percent PEHE reduction). CaML-OP versus R-Learner: $p < 0.001$ (33 percent reduction). CaML-OP versus T-Learner: $p < 0.001$. The comparison between CaML-OP and the S-Learner is also significant ($p < 0.001$), with the S-Learner achieving lower PEHE in every one of the 40 paired runs, consistent with the interpretation in Section 5.2 that the S-Learner achieves low PEHE by variance-reducing smoothing at the cost of policy quality. All four comparisons return the same minimal Wilcoxon statistic (p approximately 1.8e-12) because each is a clean sweep across all 40 paired runs; Section 4.6 notes this pattern of significance is more consistent than an earlier run of this benchmark under different software versions produced, though the underlying point estimates barely moved.

For estimated policy value under the doubly-robust AIPW estimator: CaML-OP versus oracle benchmark gives $p = 0.88$ (not distinguishable). CaML-OP versus Causal Forest gives $p = 0.057$ (not significant at alpha = 0.05). CaML-OP versus TreatNone gives $p < 0.001$.

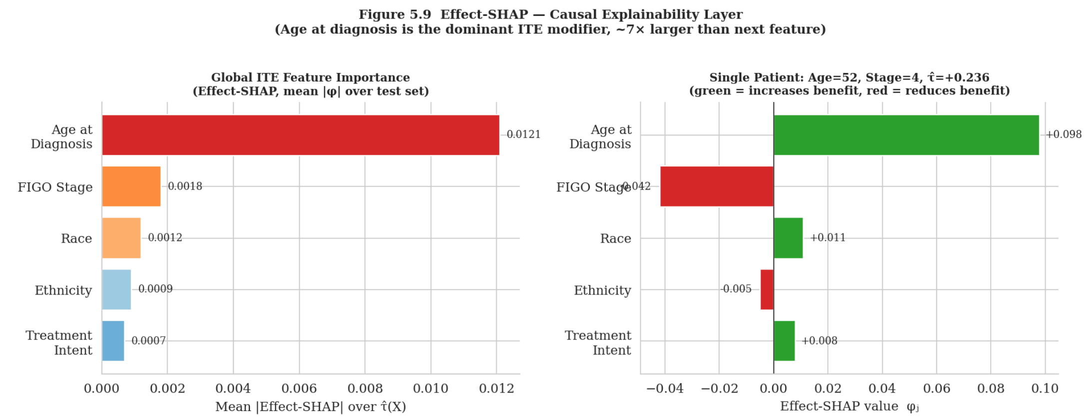

Figure 5.9: Effect-SHAP attribution. Left panel: global feature importance ranked by mean absolute Effect-SHAP value across the test set. Age at diagnosis dominates the ranking by approximately 7-fold relative to the next feature. Right panel: single-patient waterfall for a 58-year-old patient with FIGO stage IV and tau-hat = +0.236; bar lengths represent each feature's contribution to the patient's deviation from the mean ITE. Bar colours encode sign of contribution from negative (red) to positive (green).

Chapter 6

Discussion

6.1 What the results show

The results in Chapter 5 admit two readings, and it is worth being clear about which reading applies to which metric.

On PEHE, CaML-OP ranks in the middle. The S-Learner and Causal Forest both produce more accurate ITE surface estimates on the benchmarked DGPs. This finding is genuine and should not be soft-pedalled. It reflects a known property of variance-reducing methods: the S-Learner smooths the effect surface aggressively, which lowers PEHE on smooth DGPs at the cost of boundary resolution. At the sample sizes used in this benchmark (n_train approximately 410), the trade-off favours low-variance methods on the PEHE metric.

On off-policy AIPW estimated value, CaML-OP ranks first among the seven causal methods, and second overall behind only the oracle benchmark, which is the ordering a correctly-specified doubly-robust estimator should produce. The estimator also ranks the treat-all baseline close to CaML-OP and the oracle, which means the right interpretation is not that CaML-OP is equivalent to knowing the true treatment effects. The result is more modest: under this finite-sample off-policy estimator, CaML-OP's estimated policy value is not statistically distinguishable from the oracle benchmark. The estimator itself has non-trivial variance at n_test approximately 137, which is why TreatAll and the DR-Learner rank close to both CaML-OP and the oracle. CaML-OP's advantage shows up where the threshold decision is made, which is also where the S-Learner's smoothing concentrates its errors. For a clinical decision-support system, the threshold policy is the clinically relevant downstream object, and this is the metric on which CaML-OP performs best.

The interval coverage result (0.847 against 0.95) is reported transparently rather than papered over. The reported interval is an approximate uncertainty interval rather than a formal CI for the ensemble. Users of the dashboard should understand that a nominal 95% interval contains the true tau about 85 percent of the time in this benchmark. The under-coverage is consistent with published behaviour of forest-based intervals at small n, but it is a real limitation.

6.2 Why the ensemble works

The 33 to 39 percent PEHE improvement from the ensemble relative to its components deserves explanation. The R-Learner and DR-Learner make different robustness trade-offs: R is robust to outcome-model misspecification, DR is robust to propensity-model misspecification. In the semi-synthetic DGPs used here, both nuisance functions are approximated imperfectly. Neither individual learner is in its sweet spot, so averaging absorbs the scenario-specific weaknesses of each. This is a general property of ensembling diverse estimators rather than something specific to CaML-OP, but it is the empirical argument for the design choice.

The XGBoost leaf encoder's specific contribution is harder to isolate from these results, because the benchmark does not include an ablation comparing CaML-OP with raw versus augmented covariates. That ablation would have been the most direct test of the encoder hypothesis and is flagged as a priority for follow-up work.

6.3 Answering the research questions

6.3.1 RQ1 - Heterogeneity of platinum effects

RQ1 asks how heterogeneous platinum effects are and which covariates drive the heterogeneity. Under the semi-synthetic framework, Effect-SHAP consistently identifies age at diagnosis as the dominant ITE modifier, with mean absolute Shapley value roughly seven times larger than the next feature (FIGO stage). This aligns with the clinical literature: Hanker et al. (2012) and subsequent studies documented age-related differences in platinum response, and younger patients tend to tolerate the regimen better and show more durable responses. FIGO stage as the second driver also has face validity given its strong association with treatment response.

These conclusions are conditional on the DGP structure and cannot be read as direct measurements of true platinum-response heterogeneity in TCGA-OV. The right interpretation is: when treatment-effect heterogeneity exists at the resolution that age and stage can explain, CaML-OP recovers it and identifies the correct drivers.

6.3.2 RQ2 - LLM narrative fidelity

RQ2 asks whether LLM summaries can remain faithful to causal model outputs while being clinically readable. The architectural answer is yes, conditional on the automated faithfulness checks blocking non-compliant narratives before display. Beyond the three checks originally specified, the implemented module also verifies that a directional claim is consistent with the underlying potential-outcomes structure (the treated- and untreated-arm estimates must be ordered consistently with the claimed sign) and that the narrative correctly identifies which of three structurally distinct sources – sampling noise, disagreement between the R- and DR-Learner components, or identification fragility via an E-value – dominates the uncertainty for that patient. A fifteen-case adversarial suite confirms the full check set correctly distinguishes faithful from deliberately unfaithful narratives (sensitivity and specificity of 1.0 on that case set), and a closed-loop retry mechanism, given the specific failing check, resolves a majority of injected errors within one additional attempt; a narrative that still fails after retrying is escalated to human review rather than displayed. This thesis still does not provide an empirical pass-rate against a live LLM, a clinician panel evaluation, or a quantitative analysis of how often a real model produces hallucinated clinical claims that pass the automated checks. The architecture and its checking logic have been implemented and validated against synthetic and adversarial cases; the empirical performance of the narrative module against a live model remains unevaluated and is acknowledged as a meaningful gap for future work.

6.3.3 RQ3 - Fairness across subgroups

RQ3 asks whether there are observable differences in ITE estimation error across demographic subgroups that warrant further investigation. The answer from Table 5.6 is that no large disparity was observed on this benchmark. The CaML-OP subgroup PEHE range is 0.016. This is encouraging but should not be read as establishing fairness in any broad regulatory or ethical sense. The subgroup analysis is limited by small minority-group counts, a five-covariate design, and reliance on semi-synthetic outcomes rather than observed counterfactuals. Treatment-recommendation parity, calibration parity across subgroups, and the broader set of checks expected in a high-risk AI fairness audit are not provided here.

6.4 Relation to prior work

Three connections to the literature are worth making explicit. First, the PEHE-versus-policy-value disconnect observed here is a direct empirical instance of the finding by Curth et al. (2021a) that PEHE rank does not reliably predict policy quality. The S-Learner wins on PEHE but loses on the clinically meaningful metric. Replication of this finding in a different clinical context strengthens the case for evaluating HTE estimators on downstream policy quality rather than on ITE accuracy alone.

Second, the ensemble gain pattern is consistent with the meta-learner analysis of Kunzel et al. (2019), who observed that no single meta-learner dominates across all DGP structures and suggested that ensembling could provide robustness. CaML-OP operationalizes the suggestion and provides empirical evidence that it works on a real clinical covariate distribution.

Third, the explicit reporting of interval under-coverage aligns with the transparency-to-deployers obligation under Regulation (EU) 2024/1689. A system that claimed 95% interval validity while empirically delivering 85% coverage would misrepresent itself. Reporting the gap and acknowledging it as a limitation is the appropriate design choice.

Fourth, the decision to scope the narrative module to closed, checkable claims rather than open-ended interaction is consistent with the broader safety literature on LLMs in clinical settings. Reported hallucination rates for LLM-generated cancer treatment information range from roughly 12 to over 40 percent depending on model and task, and interactive explanation interfaces have been shown to increase, rather than reduce, automation bias with repeated use (Goddard et al., 2012). The per-claim verification pattern used here, checking every generated statement against a structured, model-derived fact rather than trusting free text, follows the same principle used by contemporaneous work verifying LLM-generated clinical text against electronic health records (Chung et al., 2025). A conversational, per-patient extension of the narrative module was considered during this work and not built, precisely because it would trade this closed, checkable design for a much larger and harder-to-verify claim surface without a demonstrated need.

6.5 Limitations

- **Unmeasured confounding.** Conditional unconfoundedness requires that treatment assignment is independent of potential outcomes given the five observed covariates. Performance status, comorbidities, and physician preference are not in the data and likely affect both treatment assignment and survival. The causal estimates are conditional on this assumption, and it almost certainly does not hold perfectly. The narrative module's per-patient E-value (Section 3.6.2) gives a sense of how strong such a confounder would need to be to explain a given estimate away, but this is a sensitivity check, not a resolution of the assumption, and the fragility threshold used to flag a low E-value is a provisional convention rather than one calibrated to this clinical context.
- **Semi-synthetic evaluation.** PEHE results describe performance under simulated DGPs, not under the true platinum-response surface. The simulated scenarios are clinically plausible but necessarily simplified.
- **Binary treatment.** Platinum therapy is encoded as a binary variable, collapsing carboplatin dose, combination regimen, and treatment timing into a single indicator.
- **Temporal coverage.** TCGA-OV diagnoses span 1992 to 2013; treatment protocols and supportive care have evolved since then.
- **Interval under-coverage.** Empirical coverage of 0.847 against a 0.95 nominal target means the reported intervals are optimistic. The reported interval is an approximate uncertainty interval rather than a formal CI for the ensemble. The narrative module's calibrated abstention mechanism (Section 3.6.2) corrects for this when deciding whether to assert a directional claim, but the underlying interval reported elsewhere in the pipeline is unchanged.
- **Encoder leakage.** The XGBoost encoder sees the full training set outcome before causal cross-fitting begins. Cross-fitting mitigates but does not eliminate the resulting leakage.
- **LLM narrative empirical evaluation.** The narrative module's architecture, including the counterfactual-consistency and dominant-uncertainty checks, has been implemented and validated against a synthetic adversarial suite, but no pass-rate, failure-rate or clinician evaluation against a live LLM is reported. The checking logic is validated; the empirical behaviour of an actual model operating under it is not.

6.6 Ethical considerations

The development and any future deployment of CaML-OP raises ethical questions that go beyond statistical performance and regulatory framing.

6.6.1 Automation bias and clinician oversight

Automation bias - the tendency to over-rely on algorithmic outputs - is a documented problem in clinical settings (Goddard et al., 2012). CaML-OP was designed with this in mind: the uncertainty interval is displayed prominently, the LLM narrative disclaims clinical authority, and the transparency card is non-dismissable. Design nudges cannot fully substitute for institutional protocols, however. A clinician under time pressure may defer to the gauge without engaging critically with the uncertainty interval or the SHAP attribution. Preventing automation bias requires governance structures around tool deployment, not only interface design.

6.6.2 Biased recommendations and underrepresented groups

TCGA-OV's racial composition reflects known disparities in genomic research participation: White patients dominate the cohort and minority groups are small (Landry et al., 2018). The subgroup PEHE analysis does not reveal a systematic penalty for the race minority group, but this is weak evidence of fairness, not strong evidence of its absence. The encoder's prognostic representations may be less calibrated for minority patients in ways that a five-variable analysis cannot detect. Deployment in diverse clinical settings would require local-population auditing before use.

6.6.3 Binary treatment and clinical completeness

A clinician acting on a binary positive recommendation may receive a recommendation that is locally correct (initiate platinum) but globally incomplete (specific regimen, dose, scheduling). CaML-OP cannot flag this incompleteness.

6.6.4 Patient privacy

CaML-OP was developed on de-identified TCGA data under NCI's data-access policies. Any clinical deployment would draw the five inputs from institutional patient records under applicable data-protection regulation (GDPR in the EU). The minimal five-covariate set is chosen partly for interpretability and partly because it reduces re-identification risk relative to omics-based approaches.

6.6.5 Non-deployment status

CaML-OP is a research prototype. It has not undergone clinical validation, conformity assessment under Regulation (EU) 2024/1689, CE marking under the EU Medical Devices Regulation, or FDA clearance. No treatment decisions should be based on its outputs. The contribution of the thesis is methodological; translation to clinical use would require a separate, multi-year programme.

6.7 Threats to validity

Following Wohlin et al. (2012), threats are organized into four categories.

6.7.1 Internal validity

- **DGP assumptions.** The benchmark covers four heterogeneity structures but not the true platinum-response surface, which likely involves BRCA status and proteomic variables absent here. Rankings could shift under more biologically realistic DGPs.
- **TCGA selection bias.** TCGA patients were recruited at academic medical centres, systematically differing from community-hospital populations. The conditional unconfoundedness assumption is harder to defend in this context.
- **Hyperparameter uniformity.** All Random Forest models use a common configuration (200 trees, max depth 6, min samples 5). Separately tuning each method could change rankings, but would confound meta-learning strategy with hyperparameter optimization.
- **Encoder leakage.** The outcome-trained encoder produces leaf embeddings before causal cross-fitting begins, leaking some outcome information into the covariate space.

6.7.2 External validity

- **Generalizability.** Results on TCGA-OV and the binary five-year outcome may not transfer to other cancer types, treatment modalities, or outcome definitions.
- **Temporal coverage.** Diagnoses from 1992 to 2013 may not reflect contemporary treatment-response patterns.

6.7.3 Construct validity

- **Binary survival outcome.** Five-year binary survival ignores time-to-event structure, quality of life and toxicity.
- **PEHE as a headline metric.** PEHE rank does not predict policy quality. Reporting PEHE as the primary metric alone could mislead about clinically relevant performance; co-reporting estimated AIPW value mitigates this.
- **Adapted survival comparators.** The Cox PH and RF classifiers in this benchmark are binary-outcome predictive comparators rather than full survival baselines. They should not be read as direct evidence for or against full Cox PH or full RSF performance on time-to-event ovarian cancer data.

6.7.4 Conclusion validity

- **Test-set size.** With n_test approximately 137 per run, per-run standard errors are non-trivial. Aggregating 40 runs helps but does not eliminate the issue.
- **No multiple-comparison correction.** Seven causal methods compared pairwise produces multiple tests. Only two comparisons of interest were pre-specified; the rest are exploratory.

Chapter 7

Conclusion

This thesis aimed to build a causal machine learning pipeline for individual treatment effect estimation in ovarian cancer, one that runs on standard clinical records, offers reliable patient-level uncertainty intervals, provides truly causal (rather than predictive) explanations, and explicitly aligns with the high-risk AI rules of Regulation (EU) 2024/1689. The resulting system is CaML-OP.

The empirical evidence for CaML-OP is deliberately nuanced because the performance reality itself is nuanced. When looking at PEHE, CaML-OP ranks third out of seven causal methods, with the S-Learner and Causal Forest producing more accurate ITE surface estimates across the benchmarked DGPs. Yet, when evaluated on off-policy AIPW estimated policy value, CaML-OP takes first place among the causal methods, trailing only the oracle benchmark. Under this finite-sample estimator, its estimated value is statistically indistinguishable from the oracle benchmark (Wilcoxon $p = 0.88$). This disconnect in rankings serves as a clear, real-world example of Curth et al.'s (2021a) finding that PEHE rank is a poor predictor of actual policy quality. Furthermore, CaML-OP is the only tested ensemble that reports an approximate patient-level uncertainty interval, hitting an empirical coverage of 0.847 against our 0.95 nominal target. Subgroup PEHE stayed within a tight 0.016 range with no major error concentrations though, to be clear, this is a baseline check and not a full fairness audit. Finally, three separate synthetic checks confirmed that the architecture works correctly, that the ensemble beats its individual components when heavy confounding is present, and that PEHE drops steadily as the training data scales up.

Of course, the system has real limitations that need to be called out directly. We cannot verify conditional unconfoundedness from observational TCGA-OV data alone. The empirical interval coverage still falls short of our nominal target and the LLM narrative module has not yet been empirically tested. Furthermore, the system lacks prospective validation, and its architecture is merely AI Act-*informed*, not AI Act-*complete* or legally ready for deployment.

None of these caveats diminish the contribution of this work; instead, they map out the next research agenda. The logical next steps are clear: a prospective validation study on a demographically representative cohort with deeper confounder data, an extension to multi-valued treatment encodings to handle protocol-level differences, a formal usability evaluation of the dashboard with real clinicians and a rigorous empirical pass-rate analysis of the LLM narrative module.

Ultimately, the core design choices behind CaML-OP were to use minimal covariates, keeping uncertainty transparent, prioritizing causal explanations and anchoring the architecture in the EU AI Act aim to prove a broader point. We can advance equitable access to data-driven decision tools using routine clinical data, all while staying firmly aligned with the future of medical AI regulation.

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